

House Price Prediction using Machine Learning

Project Overview

Objective

The objective of this project was to study the housing data and create machine learning models that will help predict house prices based on different property characteristics. It is also essential to find which characteristics influence the pricing of housing the most.

Dataset Overview

The Housing dataset includes data about residential properties and its features, for example, area, number of bedrooms, bathrooms, stories, parking, furnished or not, air conditioning, and other. The target variable used for predictions in the dataset is the house price.

Data Processing

Exploration of the dataset was done by means of Pandas and checks for duplicates and missing values were done. All categorical variables were encoded using one-hot encoding in order to use them for machine learning models. The dataset was split in the training and test datasets in the 80/20 ratio.

Models Used

Two machine learning models were built and tested:

- Linear Regression
- Random Forest Regressor

Results

Linear Regression model scored MAE= 970,043; RMSE = 1,324,507; $R^2 = 0.653$. Random Forest Regressor scored $R^2 = 0.613$. The Linear Regression model was more efficient according to the evaluation criteria, hence it was chosen for predicting house prices.

Key Takeaways

It was found that the most influential factors that affect the housing prices are the area, number of bathrooms, parking, and air conditioning. The larger the housing and the better its amenities the higher property value it has. A fascinating result was obtained that the Linear Regression model works better than Random Forest. That means the relation between the features and prices of housing is almost linear.

Recommendation

When selling residential properties real estate businesses should emphasize their size, additional bathrooms, parking lots, and amenities. They significantly affect the house prices.